

Parsing JSON in JavaScript

JSON (JavaScript Object Notation) is a lightweight data format used for exchanging data between a server and a client. JavaScript provides built-in methods to **parse** and **stringify** JSON.

What is JSON?

JSON is a **string representation** of data structured in key-value pairs.

Example JSON string:

```
{
  "name": "Harry",
  "age": 25,
  "skills": ["JavaScript", "Python"]
}
```

This looks like a JavaScript object, but it's actually a string.

Parsing JSON with `JSON.parse()`

To convert a JSON string into a JavaScript object, use `JSON.parse()` :

```
const jsonString = '{"name":"Harry","age":25}';

const user = JSON.parse(jsonString);
console.log(user.name); // Output: Harry
```

Note:

- The string **must be valid JSON**, otherwise `JSON.parse()` will throw an error.
 - All keys must be enclosed in double quotes (`"`).
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Stringifying JavaScript Objects with `JSON.stringify()`

To convert a JavaScript object into a JSON string, use `JSON.stringify()` :

```
const user = {  
  name: 'Harry',  
  age: 25  
};  
  
const json = JSON.stringify(user);  
console.log(json); // Output: {"name": "Harry", "age": 25}
```

Common Use Case: `localStorage`

Since `localStorage` can only store strings, you often use JSON methods to store and retrieve objects:

```
// Store object  
localStorage.setItem('user', JSON.stringify(user));  
  
// Retrieve and parse object  
const storedUser = JSON.parse(localStorage.getItem('user'));
```

Handling Errors

Wrap parsing in a `try...catch` block to handle invalid JSON:

```
try {  
  const data = JSON.parse(badJSONString);  
} catch (error) {  
  console.error('Invalid JSON:', error.message);  
}
```

Summary

- Use `JSON.parse()` to convert a JSON string into a JavaScript object.
- Use `JSON.stringify()` to convert a JavaScript object into a JSON string.
- These methods are commonly used with `localStorage`, APIs, and data exchange.